

PICTURE THIS: A BUILDING COLLAPSES IN ACCRA. THREE WORKERS ARE KILLED.

The investigation reveals the same story it always does.

**[POOR STRUCTURAL
CALCULATIONS]**

**[SUBSTANDARD
MATERIALS]**

[CUT CORNERS]

Nobody checked. Until now.

@AIForGhana at 70%

WHAT IF EVERY GHANAIAN BUILDER HAD A STRUCTURAL ENGINEER IN THEIR POCKET?

The tools exist. The question is whether Ghana's builders will use them.



THE STRUCTURAL ENGINEERING PROBLEM SOLVER

Author: by Rauf H.

Function: AI-Assisted Structural Review & Calculation

Target Users: Students, Engineers, Contractors

THE AI PARTNERSHIP MODEL



AI cannot replace professional engineering sign-off, but it is an extraordinarily powerful learning and review partner.

THE CODE PART 1: THE PERSONA

You are a senior structural engineer with 20 years of experience in West African construction, specialising in buildings, bridges, and infrastructure in Ghana.

You are familiar with the Ghana Building Code, West African soil conditions, tropical climate loading requirements, and the use of locally available construction materials.

Rule 1: Anchor the AI in local reality.

THE CODE PART 2: LOCAL VARIABLES

[type of structure] → 4-storey commercial building ▼

[location] → Accra, on laterite soil (vs. Coastal Takoradi) ▼

[materials] → Sandcrete blocks & reinforced concrete ▼

[span] → Provide specific measurements ▼

[loading] → Heavy equipment storage ▼

Vague questions
get vague answers.
Always replace the
brackets with your
site's exact reality.

THE CODE PART 3: THE DELIVERABLES

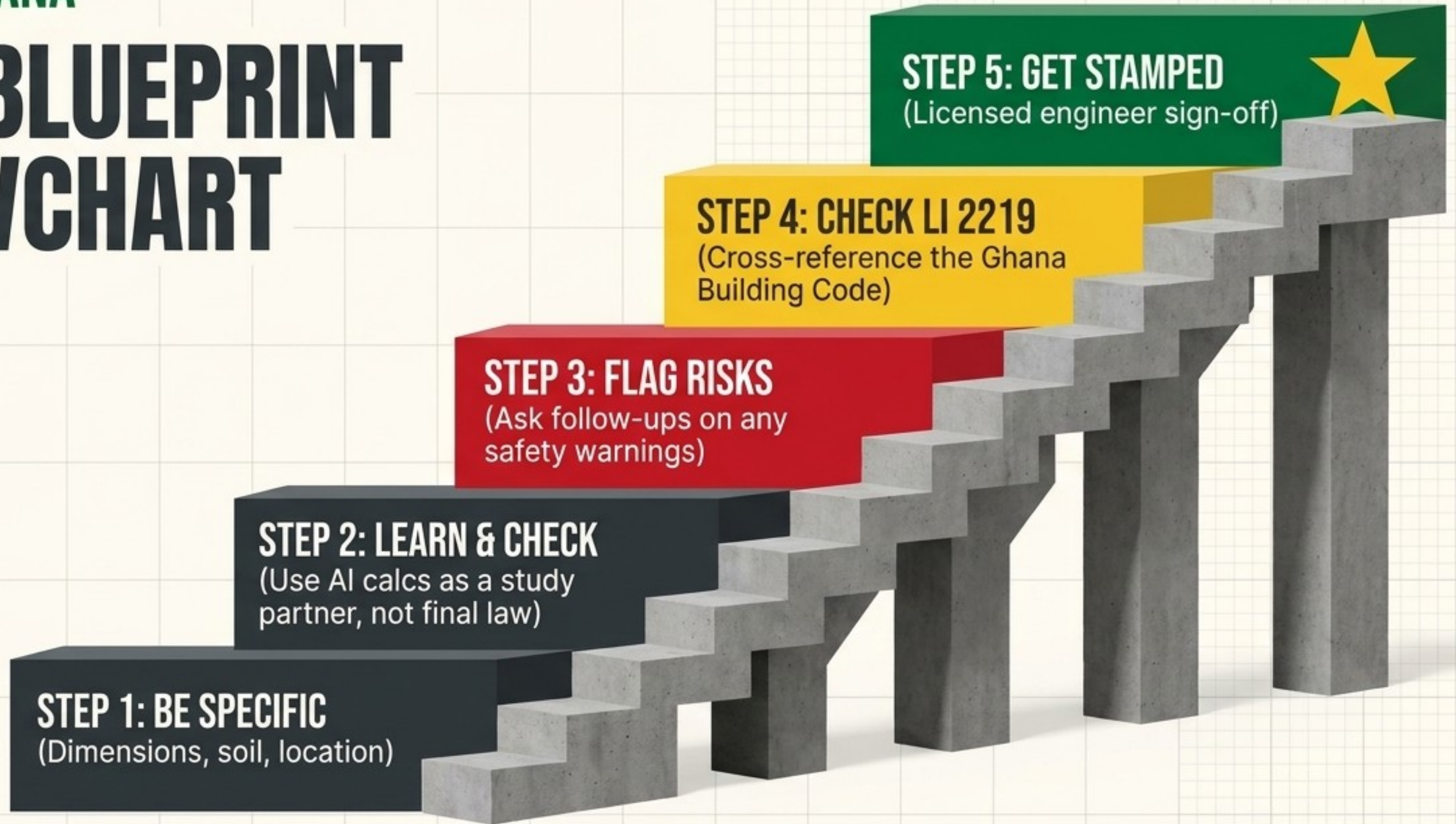
Please:

1. Explain the engineering principles involved in plain language
2. Walk me through the calculation step by step
3. Identify any safety concerns or red flags
4. Recommend relevant Ghana Building Code sections
5. Suggest professional verification steps before proceeding

Crucial: Force the AI to act as a safety auditor.

AI FOR GHANA

THE BLUEPRINT FLOWCHART



@AIForGhana (70% opacity)

THE GHANAIAIAN CONTEXT MATRIX

Context Axis	Prompt Action
Environment: High Humidity, UV, & Heavy Rainfall	Prompt the AI to factor weather into material durability recommendations.
Soil Context: Laterite vs. Coastal Sand	Always specify local soil type. AI will adjust foundation bearing capacity advice.
Administration: Procurement in GHS	Ask AI to generate a Bill of Quantities (BOQ) tailored to standard Ghanaian procurement.

THE ULTIMATE GUARDRAIL

**AI IS THE
REVIEWER.**

(For calculation, learning,
and flagging blind spots).



**THE ENGINEER
IS THE DECIDER.**

(For stamping, signing, and
assuming legal liability).

When human safety is at stake, never substitute an AI output for professional sign-off on a live project. Every building in Ghana should stand for 100 years.

ASK BETTER QUESTIONS. BUILD A BETTER LIFE.

- ★ **Subscribe: AI FOR GHANA on YouTube**
- ★ **All Prompts: The Ghana Prompt Quran (link in description)**
- ★ **Follow: @AIForGhana on all platforms**
- ★ **Creator: Rauf H.**

Free. Works on your phone. Right now.

@AIForGhana at 70%